Relating the dependent and ordinary actions of a function on a path: apd of f along p equals transport of f(x) concatenated with ap of f

The Univalent Foundations Program

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Editorial note. The counted statement is the author lemma labelled thm:apd-const (source lines 924-927); the proof span 928-936 is the author's first proof environment following it, retained line for line.

Editorial note. The transconst notation used by the proof is defined by the retained context (the statement of the transport-in-constant-families lemma).

Editorial note. The author's second proof at the following lines is not counted and not retained on the card face.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the coherence between the dependent action on paths and the ordinary action composed with transport in a constant family. Only the counted lemma and its first author proof are retained; the author's second proof remains uncounted in the source.

Allowed standard prerequisites (editorial).

- Dependent type families over A, sections f of such families, and the dependent action of f on paths (apd), with its definitional behaviour on refl.
- The ordinary action of a function on paths (ap) with its definitional behaviour on refl.
- Transport along a path and the transconst notation for transport in a constant family, both introduced by the retained context.
- Path concatenation and inversion with their definitional behaviour on refl.

Lemma 0.0.1. *If $P : A \to \mathcal{U}$ is defined by $P(x) :\equiv B$ for a fixed $B : \mathcal{U}$, then for any $x, y : A$ and $p : x = y$ and $b : B$ we have a path*

$$\mathsf{transportconst}_p^B(b) : \mathsf{transport}^P(p, b) = b.$$

Lemma 0.0.2. *For $f : A \to B$ and $p : x =_A y$, we have*

$$\mathsf{apd}_f(p) = \mathsf{transportconst}_p^B(f(x)) \cdot \mathsf{ap}_f(p).$$

First proof. Let $D : \prod_{(x,y:A)}(x = y) \to \mathcal{U}$ be the type family defined by

$$D(x, y, p) :\equiv \big(\mathsf{apd}_f(p) = \mathsf{transportconst}^B_p(f(x)) \cdot \mathsf{ap}_f(p)\big).$$

Thus, we have

$$D(x, x, \mathsf{refl}_x) \equiv \big(\mathsf{apd}_f(\mathsf{refl}_x) = \mathsf{transportconst}^B_{\mathsf{refl}_x}(f(x)) \cdot \mathsf{ap}_f(\mathsf{refl}_x)\big).$$

But by definition, all three paths appearing in this type are $\mathsf{refl}_{f(x)}$, so we have

$$\mathsf{refl}_{\mathsf{refl}_{f(x)}} : D(x, x, \mathsf{refl}_x).$$

Thus, path induction gives us an element of $\prod_{(x,y:A)} \prod_{(p:x=y)} D(x, y, p)$, which is what we wanted. □